Linear systems of three unknowns

1

a No **b** Yes **c** Yes **d** No

e No **f** No

2

a $x = 1, y = 3, z = -2$

b $x = 1,\ y = 5,\ z = 8$

c $x = 4, y = -5, z = -2$

d $x = -3, y = 5, z = -2$

e $x = -3, y = 1, z = 2$

f $x = 2, y = -1, z = -3$

g $x = 4, y = -1, z = 3$

h $x = 8, y = 10, z = 9$

i $x = 4, y = 8, z = 6$

j $x = 1, y = 2, z = -3$

3

a There are an infinite number of solutions.

b There are no solutions.

4

a $y = t - 1$

b $x = 2t - 11$

c $x = 2t - 11, y = t - 1, z = t$

5

a $y = -x - 2t + 9$

b $x = -11t + 34$

c $y = 9t - 25$

d $x = -11t + 34, y = 9t - 25, z = t$

6

a $x = 3y - 3t + 6$

b $y = -t - 1$

c $x = -6t + 3$

d $x = -6t + 3, y = -t - 1, z = t$

7

a No solution

b Inconsistent

8

a $x = 2, y = 5, z = 7$

b Independent.

9

a $x = 0.2, y = 0.5, z = 0.1$

b Consistent

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10 Example answer:

$x + y + z = -7$

$x + y - z = -5$

$x - y + z = -3$

11

a $x = \frac{1}{6}, y = \frac{1}{3}, z = \frac{1}{4}$

b $x = \frac{1}{3}, y = \frac{1}{6}, z = \frac{1}{2}$

12

a

i $x = 7, y = 4, z = 3$

ii Consistent

iii Independent

b

i $x = 4, y = 1, z = 6$

ii Consistent

iii Independent

c

i $x = 8, y = 8, z = 8$

ii Consistent

iii Independent

d

i $x = 19, y = 4, z = 16$

ii Consistent

iii Independent

e

i $x = 2, y = 3, z = 8$

ii Consistent

iii Independent

f

i $x = 13, y = 9, z = 1$

ii Consistent

iii Independent

g

i $x = 9, y = 7, z = 7$

ii Consistent

iii Independent

h

i $x = 3, y = 1, z = 9$

ii Consistent

iii Independent

i

i $x = 0.3, y = 0.5, z = 0.5$

ii Consistent

iii Independent

Applications

13

a $4A + 3B + 12C = 48$

b $4A + 3B + 6C = 36$

c $2A + B + C = 12$

d $A = 3,\ B = 4,\ C = 2$

14

a $x = 1,\ y = 1,\ z = -1$

b True

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15

a $\begin{aligned} a+b+c &= 15 \\ 4a+2b+c &= 5 \\ 16a+4b+c &= -3 \end{aligned}$

b $a = 2,\ b = -16,\ c = 29$

c $y = 2x^2 - 16x + 29$

16

a $\begin{aligned} 2a-4b+c &= -20 \\ 5a+b+c &= -26 \\ 3a-3b+c &= -18 \end{aligned}$

b $a = 8,\ b = -6,\ c = -60$

c $x^2 + y^2 + 8x - 6y - 60 = 0$

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